

ICD Final Report

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1. Verify DRC Result

```
VERIFY DRC ..... Sub-Area: {174.400 99.200 99.200 121.920} 24 of 25  
VERIFY DRC ..... Sub-Area : 24 complete 0 Viols.  
VERIFY DRC ..... Sub-Area: {99.200 99.200 122.670 121.920} 25 of 25  
VERIFY DRC ..... Sub-Area : 25 complete 0 Viols.
```

Verification Complete : 0 Viols.

*** End Verify DRC (CPU: 0:00:02.6 ELAPSED TIME: 3.00 MEM: 256.1M) ***

2. Verify Regular Connectivity Result

innovus 13> VERIFY_CONNECTIVITY use new engine.

***** Start: VERIFY CONNECTIVITY *****

Start Time: Sat May 30 14:56:39 2026

Design Name: core

Database Units: 2000

Design Boundary: (0.0000, 0.0000) (122.6700, 121.9200)

Error Limit = 1000; Warning Limit = 50

Check specified nets

**** 14:56:39 **** Processed 5000 nets.

**** 14:56:39 **** Processed 10000 nets.

Begin Summary

Found no problems or warnings.

End Summary

End Time: Sat May 30 14:56:40 2026

Time Elapsed: 0:00:01.0

***** End: VERIFY CONNECTIVITY *****

Verification Complete : 0 Viols. 0 Wrngs.

(CPU Time: 0:00:00.3 MEM: 0.000M)

3. Analyze Floorplan Result (Area Result)

```

= total # of pins 50699 / total area 157092.
***** Analyze Floorplan *****
Die Area(um^2)      : 14955.93
Core Area(um^2)     : 8143.65
Chip Density (Counting Std Cells and MACROs and IOs): 54.451%
Core Density (Counting Std Cells and MACROs): 100.000%
Average utilization  : 100.000%
Number of instance(s) : 30530
Number of Macro(s)    : 0
Number of IO Pin(s)   : 194
Number of Power Domain(s) : 0
***** Estimation Results *****
*****

```

4. Time Result

(1) tb1

```

*Verdi* : End of traversing.

*****
*                RESULT                *
*****

# #####
# # # # # # # # # # # # # # # # # #
# # # # # CORRECT # # # # # # # # # #
# # # # # # # # # # # # # # # # # #
# #####

-----
CONGRATULATION! ALL DATA PASS!
-----

$finish called from file "../00_TESTBED/testbench.v", line 221.
$finish at simulation time 16507500
V C S   S i m u l a t i o n   R e p o r t
Time: 16507500 ps
CPU Time: 4.860 seconds; Data structure size: 4.4Mb
Sat May 30 22:19:21 2026
CPU time: 2.585 seconds to compile + .385 seconds to elab + .714 seconds to link + 4.905 seconds in simulation

```

(2) tb2

```
*Verdi* : End of traversing.

*****
*                RESULT                *
*****

# #####
# # # * *
# # # CORRECT # |
# # # # \_/_/
# #####

-----
CONGRATULATION! ALL DATA PASS!
-----

$finish called from file "../00_TESTBED/testbench.v", line 221.
$finish at simulation time 16507500
V C S S i m u l a t i o n R e p o r t
Time: 16507500 ps
CPU Time: 5.390 seconds; Data structure size: 4.4Mb
Sat May 30 22:18:29 2026
CPU time: 2.583 seconds to compile + .390 seconds to elab + .680 seconds to link + 5.429 seconds in simulation
```

(3) tb3

```
*Verdi* : End of traversing.

*****
*                RESULT                *
*****

# #####
# # # * *
# # # CORRECT # |
# # # # \_/_/
# #####

-----
CONGRATULATION! ALL DATA PASS!
-----

$finish called from file "../00_TESTBED/testbench.v", line 221.
$finish at simulation time 16507500
V C S S i m u l a t i o n R e p o r t
Time: 16507500 ps
CPU Time: 4.960 seconds; Data structure size: 4.4Mb
Sat May 30 22:17:06 2026
CPU time: 2.503 seconds to compile + .393 seconds to elab + .685 seconds to link + 5.005 seconds in simulation
[ic_design-40@ic-design-vm-1 05_POST]$
```

(4) tb4

```

ic_design-40@ic-design-vm-1:05_POST
File Edit View Search Terminal Help
*Verdi* : End of traversing.

*****
*                RESULT                *
*****

# #####
# # # * *
# # # CORRECT # |
# # # # \_/_/
# #####

-----
CONGRATULATION! ALL DATA PASS!
-----

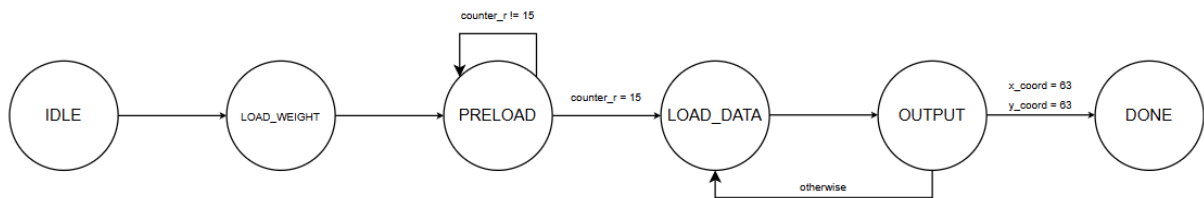
$finish called from file "../00_TESTBED/testbench.v", line 221.
$finish at simulation time 16507500
V C S Simulation Report
Time: 16507500 ps
CPU Time: 5.590 seconds; Data structure size: 4.4Mb
Sat May 30 22:15:58 2026
CPU time: 2.607 seconds to compile + .392 seconds to elab + .710 seconds to link + 5.627 seconds in simulation
ic_design-40@ic-design-vm-1:05_POST$

```

4. Explanation

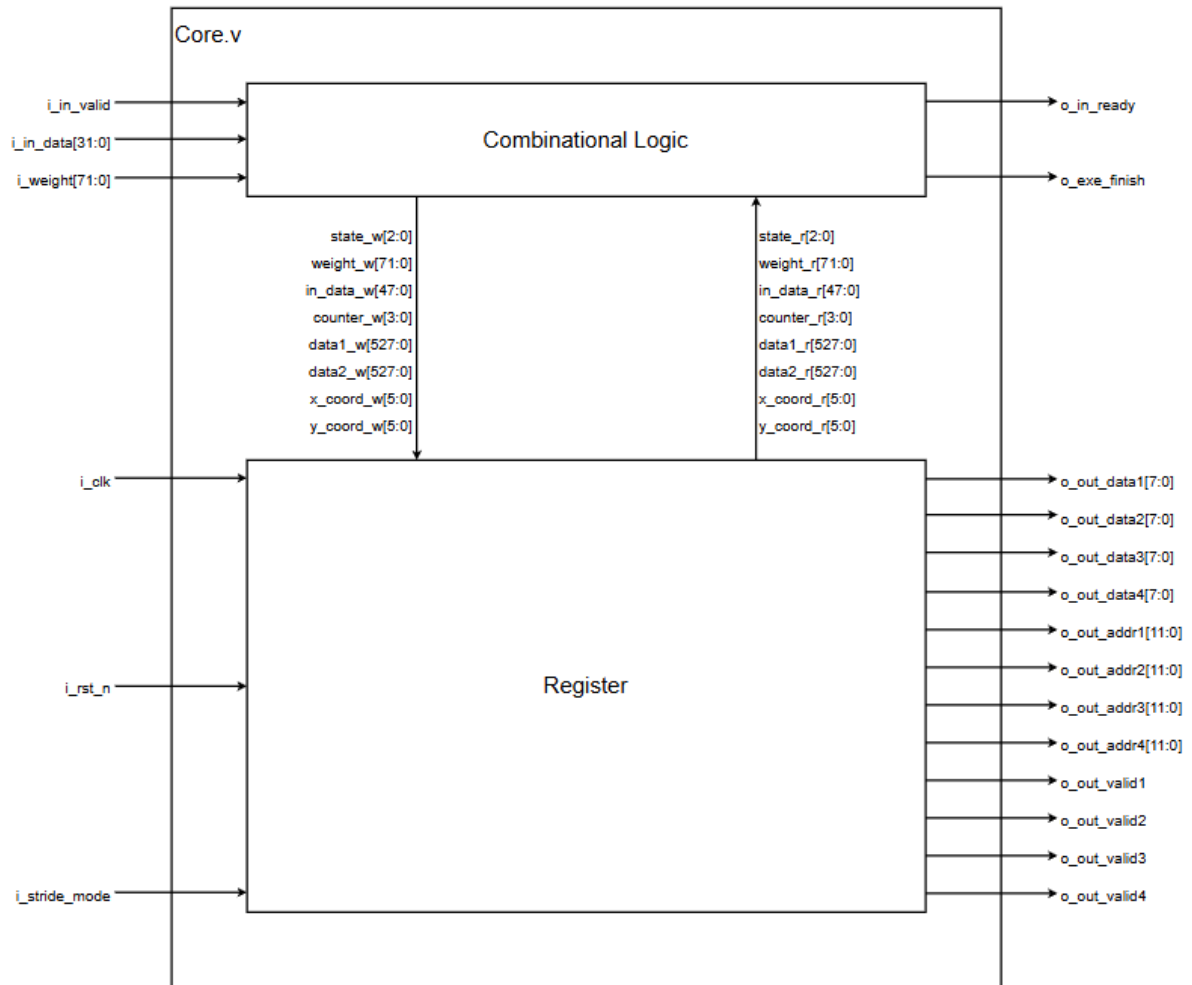
Cycle time: 7.5 ns (133MHz)

(1) State graph



State	Explanation
IDLE	Waiting for start
LOAD_WEIGHT	Load weight
PRELOAD	Load the first-row data (counter_r: calculate how many times the data is being loaded)
LOAD_DATA	Load data and calculate
OUTPUT	Output the answer (coord: identify the address that is calculated)
DONE	Done

(2) System Architecture

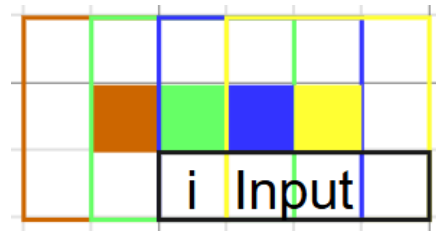


Variable	Explanation
state[2:0]	Identify the state
weight[71:0]	Store the weight from i_weight
in_data[47:0]	Store the input data for follow-up operation
counter[3:0]	Count the preload times
data1[527:0]	Store the previous row of data for convolution
data2[527:0]	Store the previous row of data for convolution
x_coord[5:0]	Store the x coordinate of the node that is calculated
y_coord[5:0]	Store the y coordinate of the node that is calculated

(3) Output explanation

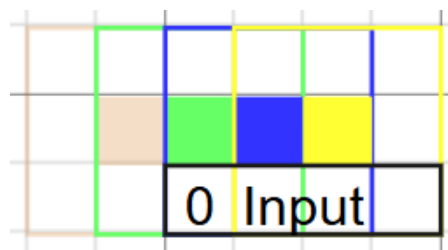
When 4 input data come in, let we call them the i -th, $(i+1)$ -th, $(i+2)$ -th, $(i+3)$ -th data, the calculation for $(i-17)$, $(i-16)$, $(i-15)$, $(i-14)$ are ready. According to the graph below, it is necessary to store 6 items in the `in_data` to perform convolution. Also, we assign `out1` to $(i-17)$, `out2` to $(i-16)$, `out3` to $(i-15)$, `out4` to $(i-14)$. Also, in stride-two situation,

only out2 and out4 can be used. However, there are some exceptions when x_coord is 0, 60, and 63, and will thus affect the data flow.



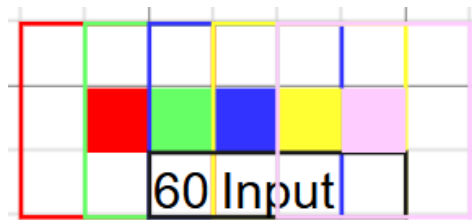
a. $x_coord = 0$

When $x_coord = 0$, out1 can't be used since it won't provide any valid output. Also, for stride-two situation, only out2 and out4 is valid.



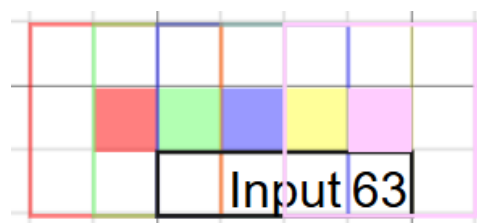
b. $x_coord = 60$

When $x_coord = 60$, there are five cells ready for outputting the result. However, there can only be four outputs. Hence, the last cell must output along, and thus x_coord then be 63 after this.



c. $x_coord = 63$

When $x_coord = 63$, there will be no input data. Also, only the 63-th cell needs to be output. We decided to let out4 to output the result.



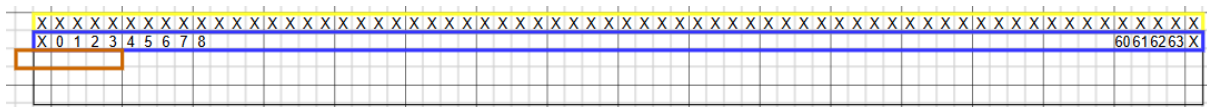
(4) Data flow explanation

To reduce the area and the time, it is necessary to think up a method rather than store all data and then calculate them. Hence, we only store two rows of data, and when the data comes in, we calculate them immediately, then output the result in the next cycle. To fulfill this implementation, data flow is important.

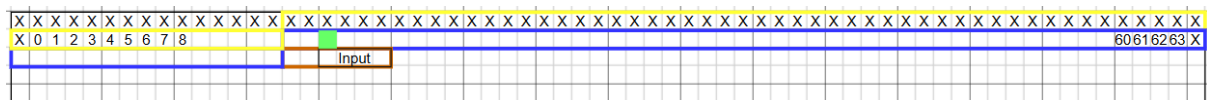
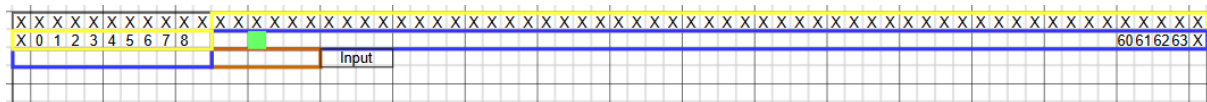
(data1 -> yellow; data2 -> blue; in_data -> red; green -> node that is calculated)

a. Initialize (Preload)

First, we need to input some data to initialize the structure. Hence, at the PRELOAD state, we read input 16 times to initialize data2.



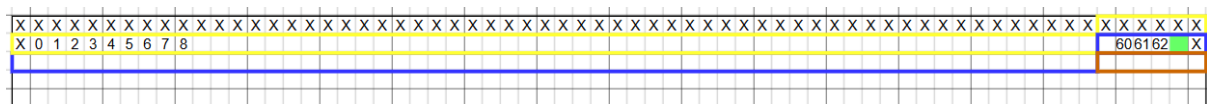
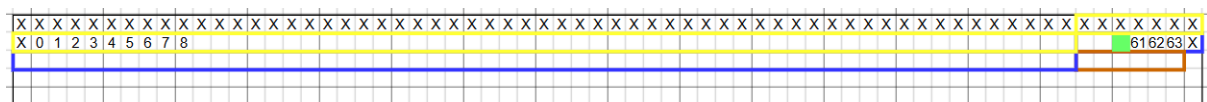
b. x_coord != 0 or x_coord != 63



The data should all be shifted by 4 cells to insert the input.

```
data1_w = {data1_r[495:0], data2_r[527:496]};
data2_w = {data2_r[495:0], in_data_r[47:16]};
in_data_w = {in_data_r[15:0], i_in_data};
```

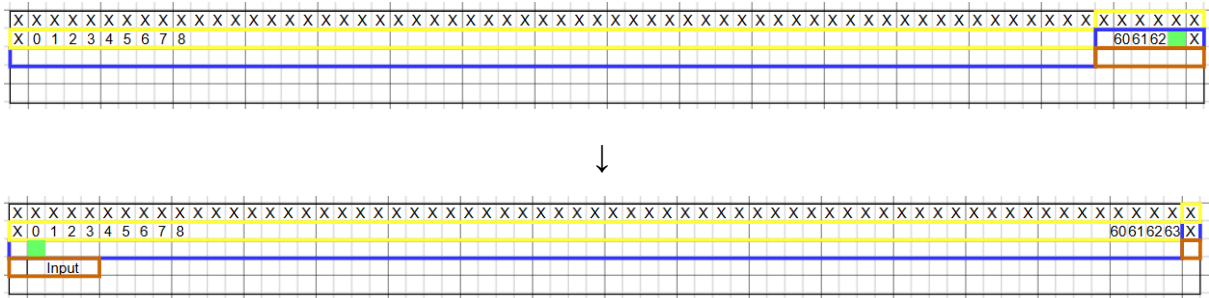
c. x_coord = 63



The data should all be shifted by 1 cell to calculate the result.

```
else if(x_coord_r == 63) begin
    data1_w = {data1_r[519:0], data2_r[527:520]};
    data2_w = {data2_r[519:0], in_data_r[47:40]};
    in_data_w = {in_data_r[39:0], 8'b0};
end
```

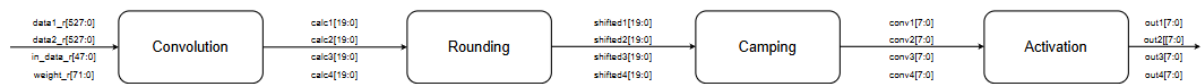
d. x_coord = 0



The data should all be shifted by 5 cells to calculate the result.

```
if (x_coord_r == 0) begin
    data1_w = {data1_r[487:0], data2_r[527:488]};
    data2_w = {data2_r[487:0], in_data_r[47:8]};
    in_data_w = {16'b0, i_in_data};
end
```

(5) Convolution & Activation



a. Convolution

Since the weights are signed numbers, the calculation is performed by using signed bits. The input bits are 8 bits * 9 bits (with a signed bit) * 9, which is up to 20 bits.

b. Rounding

The result after convolution is a number with 7 decimal digits. To round the number, we simply add 20'sd64 and then right shift for 7 digits.

c. Camping

We perform camping by using if-else conditions. Camp the negative number to 0 and those larger than 255 to 255.

d. Activation

Since the output integer is in the range of 0~40, we perform 6 times binary search in one cycle to find out the two-thirds power of the input integer.